

Main Events

Absolute Start Time	Duration	Title
00:00:00:00	00:00:05:00	McDonalds Ad
00:00:05:00	00:00:15:00	Walmart Ad
00:00:20:00	00:00:30:00	Panda Express Ad
00:00:50:00	00:00:30:00	...
00:01:20:00	00:00:30:00	...
00:01:50:00	00:00:10:00	...
00:02:00:00	00:00:15:00	...
...	...	...

Inner Events

Relative Start Time	Duration	Title	(Absolute Start Time)
00:00:00:00	00:00:01:00	Inner Event 1	00:00:20:00
00:00:05:00	00:00:05:00	Inner Event 2	00:00:25:00
00:00:06:00	00:00:10:00	Inner Event 3	00:00:26:00
...	...	...	

There is a service that maintains a 24/7 playlist of main events, where a main event is always playing.

The absolute start time of a future main event is calculated by taking the absolute start time + the duration of the previous main event. Therefore, the playlist of main events cascades without gaps or overlaps. If the Walmart ad was deleted, the Panda Express ad would shift up and the start times would re-calculate. The duration of an existing main event might be updated, and the start times of all subsequent events would re-calculate.

Inner events are scheduled within each main event with a relative start time. The absolute start time of an inner event is calculated by taking the start time of its parent main event + the relative start time of the inner event. Inner events are independent of one another. There might be gaps and overlaps of inner events. For example, between inner event 1 and 2 there is a gap of 4 seconds, and between inner event 2 and 3 there is an overlap of 4 seconds.

A playlist might have 10,000 future main events, and each main event might have 50 inner events.

Different users and systems subscribe to the playlist. Users (humans) view a playlist via a web browser. They might want to see the current event and the next 10 events. They might want to scroll quickly down the list. They might want to add/update/copy/paste/delete a main event or inner event. A system may be interested in subscribing to main and/or inner events that are relevant to their function.

Thoughts for discussion:

- Data structures that the playlist service could use
- Subscription API
 - The entire playlist could hold a lot of data. Could partial updates be a thing? If so, how would the client be assured that their representation of the platlist is consistent with that of the service?
- Things that could go wrong
 - ie, what if a user deletes the next main event 10 milliseconds before it's scheduled to start?
 - What if an inner event's relative start time falls outside of it's parent main event's duration?
 - ...